Excercise 3

Draw the symbols used in topographic sheet to show various categories of

a. Railway lines
b. Roadways
c. Water features
d. Physical features
e. Cultural features
f. Vegetation

9.2 Scale

The scale is defined as the ratio between the distance of two points on the map and their corresponding distance on the ground. The scale is an essential element in all types of maps. The scale of the map permits the user to convert distance on the map to distance on the ground.

A map scale provides the relationship between the map and the whole or a part of the earth's surface shown on it. We can also express this relationship as a ratio of distances between two points on the map and their corresponding distance on the ground. The scale can be represented as a fraction where the numerator refers to map distance and the denominator refers to ground distance.

There are at least three ways of representing scale. They are:

1. Statement Scale
2. Representative Fraction (R. F.)
3. Graphical or Bar Scale

1. Statement Scale

The scale of a map may be indicated in the form of a written statement. For example, if on a map a written statement appears as stating 1 cm represents 10 km, it means that on that map a distance of 1 cm is representing 10 km of the corresponding distance on the ground.1 inch equals 16 miles. This example tells us that 1 inch on the map represents 16 miles on the ground. This is the easiest scale to understand because it generally uses familiar units.

Example :

1 centimetre to 10 kilometres

2. Representative fraction (RF)

It shows the relationship between the map distance and the corresponding ground distance in the same units of length. R. F. is generally shown as a fraction. For example, a fraction of 1 : 40,000 shows that one unit of length on the map represents 40,000 of the same units on the ground i.e; 1 cm or 1 inch on the map represents 40,000 cm and 40,000 inches, respectively on the ground.

R.F has the following characteristics :

1. If the numerator is in centimeters, then the denominator is in metres and kilometres.
2. If the numerator is in inch, then the denominator is in miles.

Example :

RF is represented as 1/40,000 or 1:40,000

3. Graphic or bar scale

In this type of scale the map distances and the corresponding ground distances are marked using a line bar with primary and secondary divisions on it. However, unlike the statement of the scale method, the graphical scale stands valid even when the map is reduced or enlarged. This is the unique advantage of the graphical method of representing scale.

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